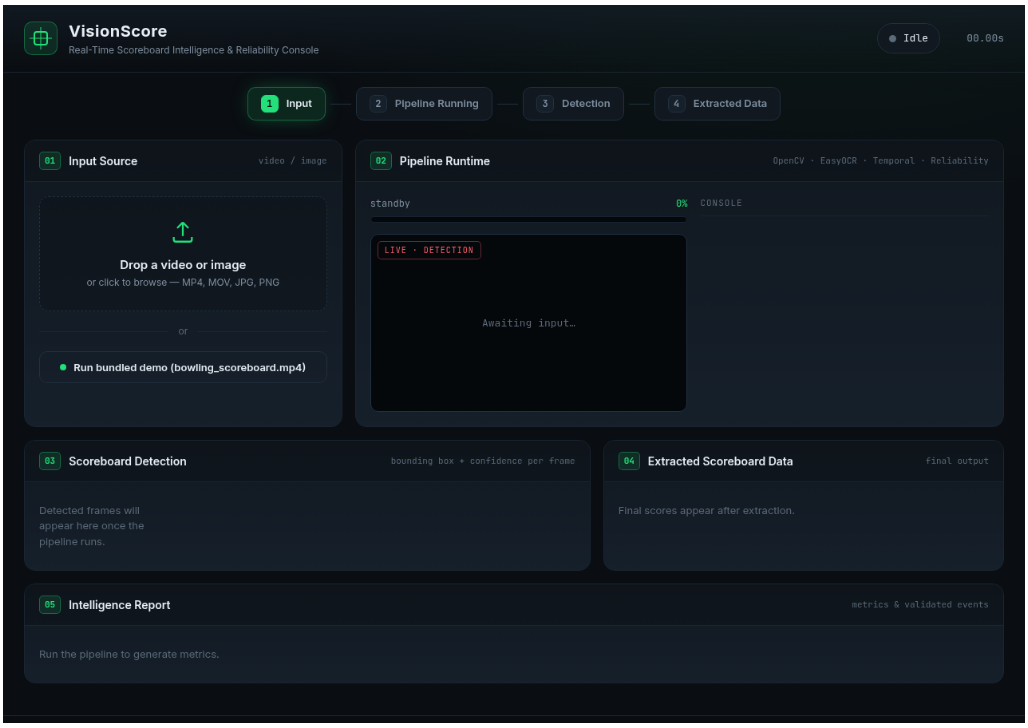
VisionScore

Real-Time Scoreboard Intelligence & Reliability System

Project documentation. This document walks through the complete working solution with screenshots of the input video, the project running, the scoreboard being detected, and the final extracted scoreboard data.

Pipeline: Video → Frame extraction → Scoreboard localization (OpenCV) → OCR (EasyOCR) → Temporal series → Reliability analysis → Reliable structured data.

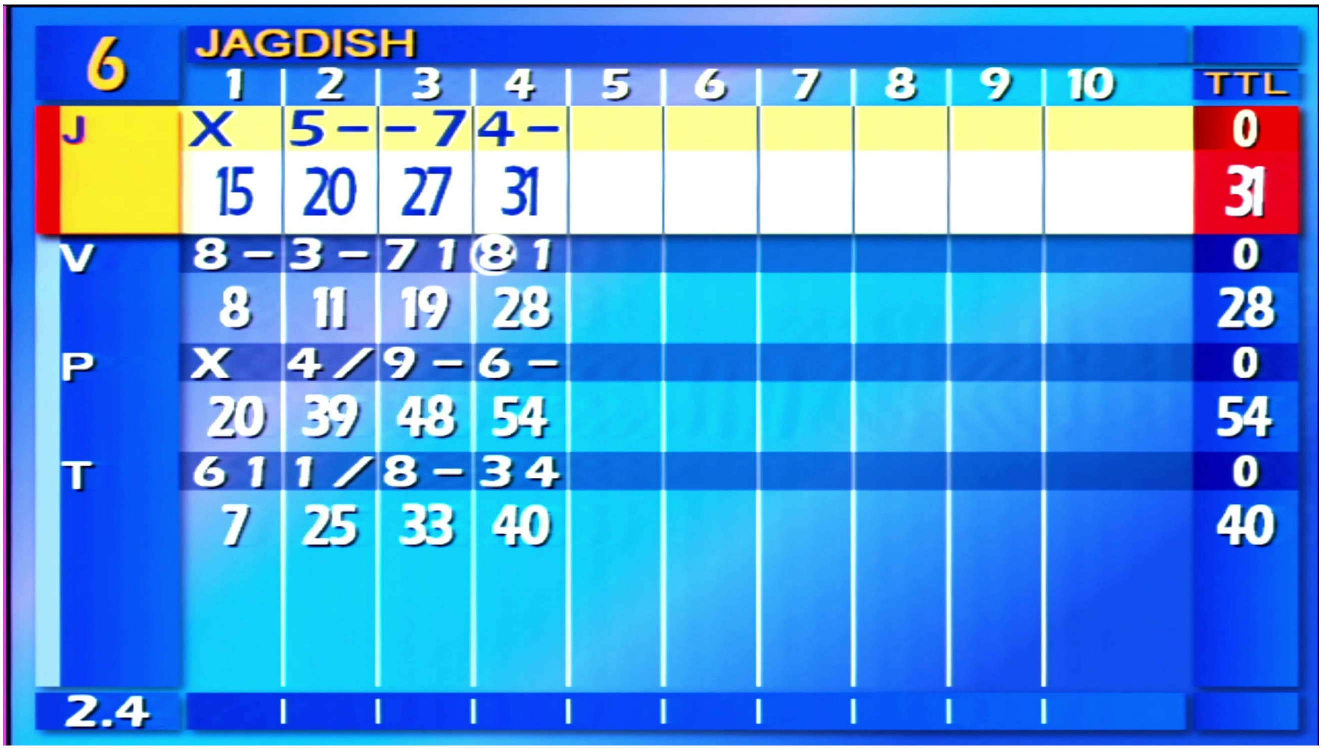
Tech stack: Python, OpenCV, NumPy, EasyOCR, Flask (dashboard).



VisionScore dashboard - the single-screen console.

1. Input Video / Frame

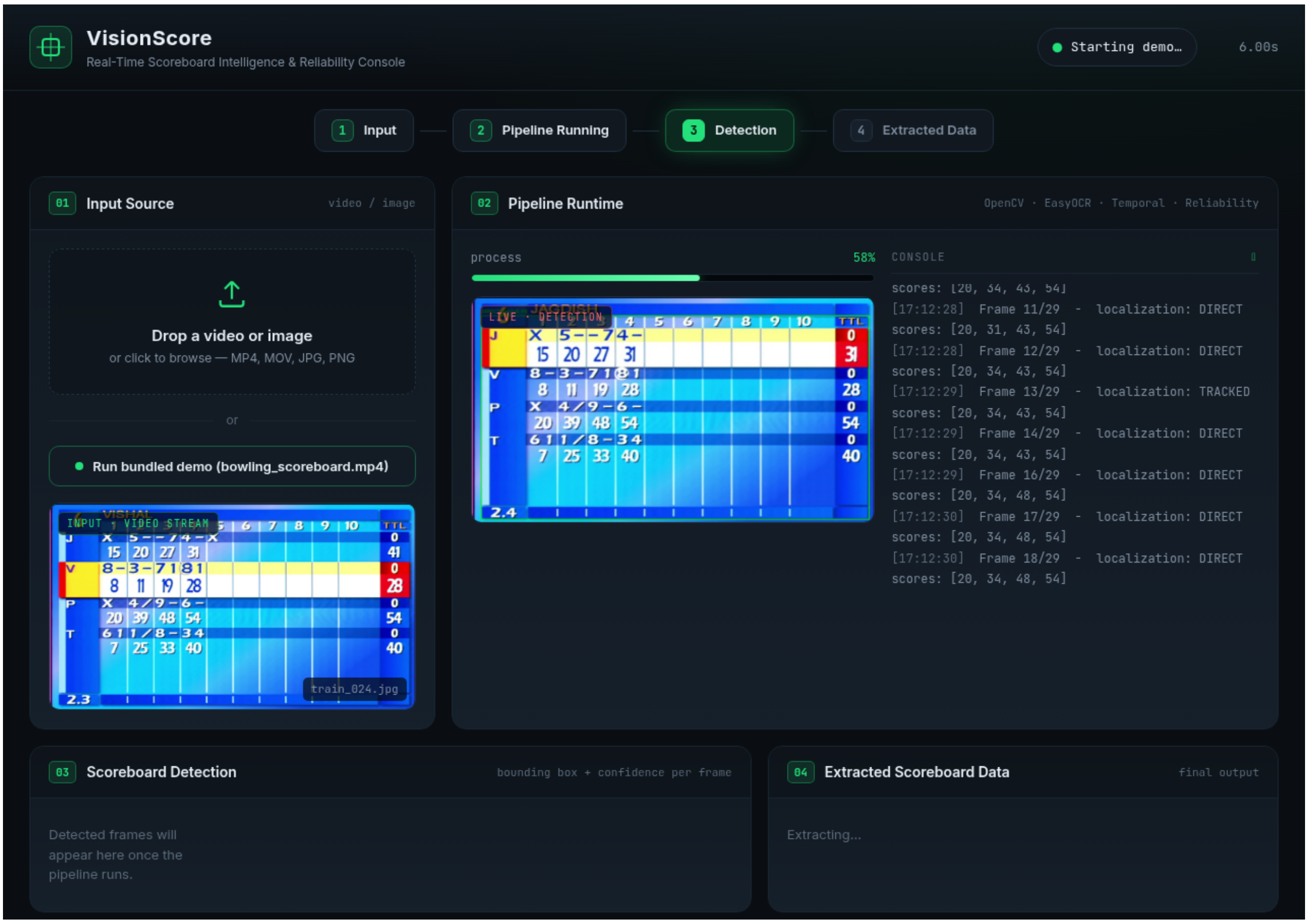
The input is a bowling broadcast clip (**bowling_scoreboard.mp4**) containing an on-screen scoreboard. The system extracts frames from this video as the starting point. Below is a representative input frame showing the four players (J, V, P, T) and their running totals in the TTL column.



A single input frame extracted from bowling_scoreboard.mp4.

2. Code / Project Running

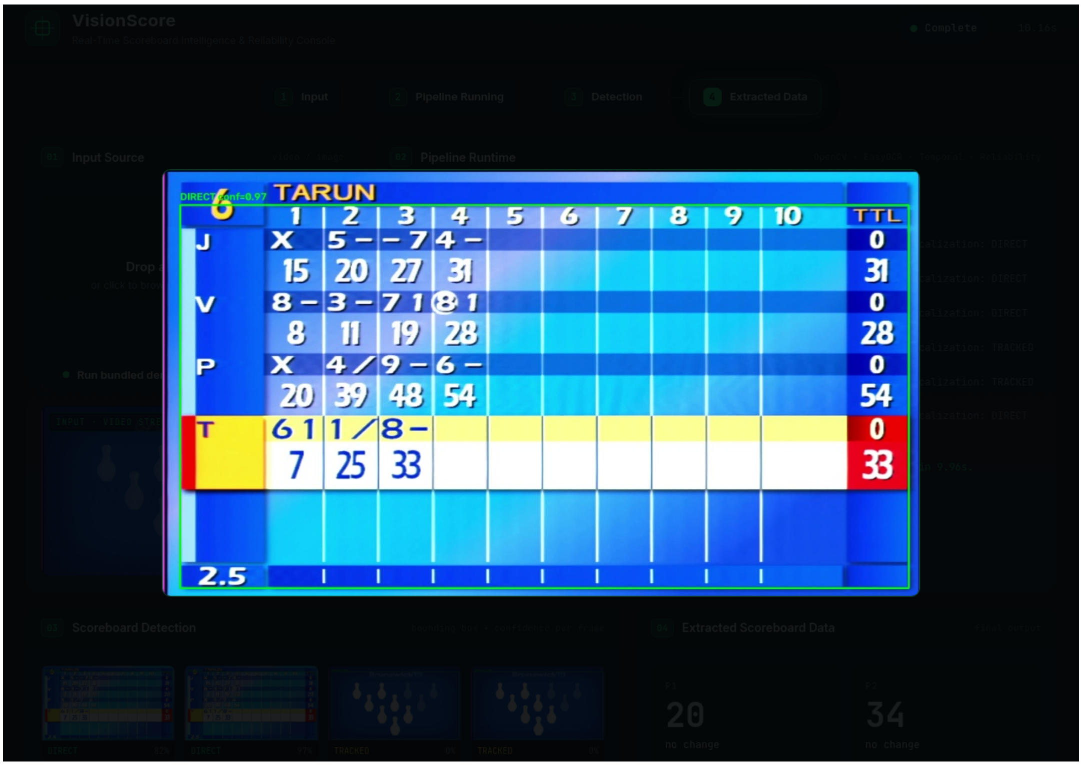
Clicking **Run bundled demo** (or uploading a file) starts the real pipeline. The dashboard streams live progress: the input video plays on the left, and the **Pipeline Runtime** console logs each frame's localization result (DIRECT / TRACKED) and the OCR-read scores in real time.



The pipeline running live - progress bar, per-frame console output, and the current detection preview.

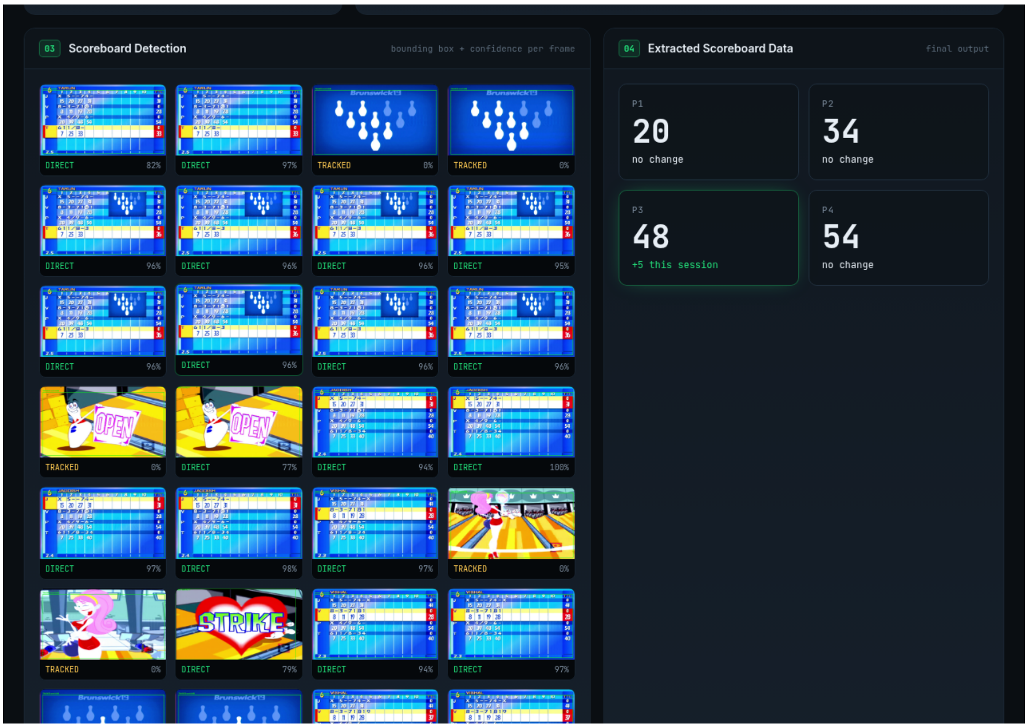
3. Scoreboard Being Detected / Extracted

Each processed frame is annotated with a green bounding box around the localized scoreboard, its detection mode (DIRECT vs. TRACKED), and a confidence score. The close-up below shows a single detection at 97% confidence.



Detected scoreboard with bounding box and confidence label.

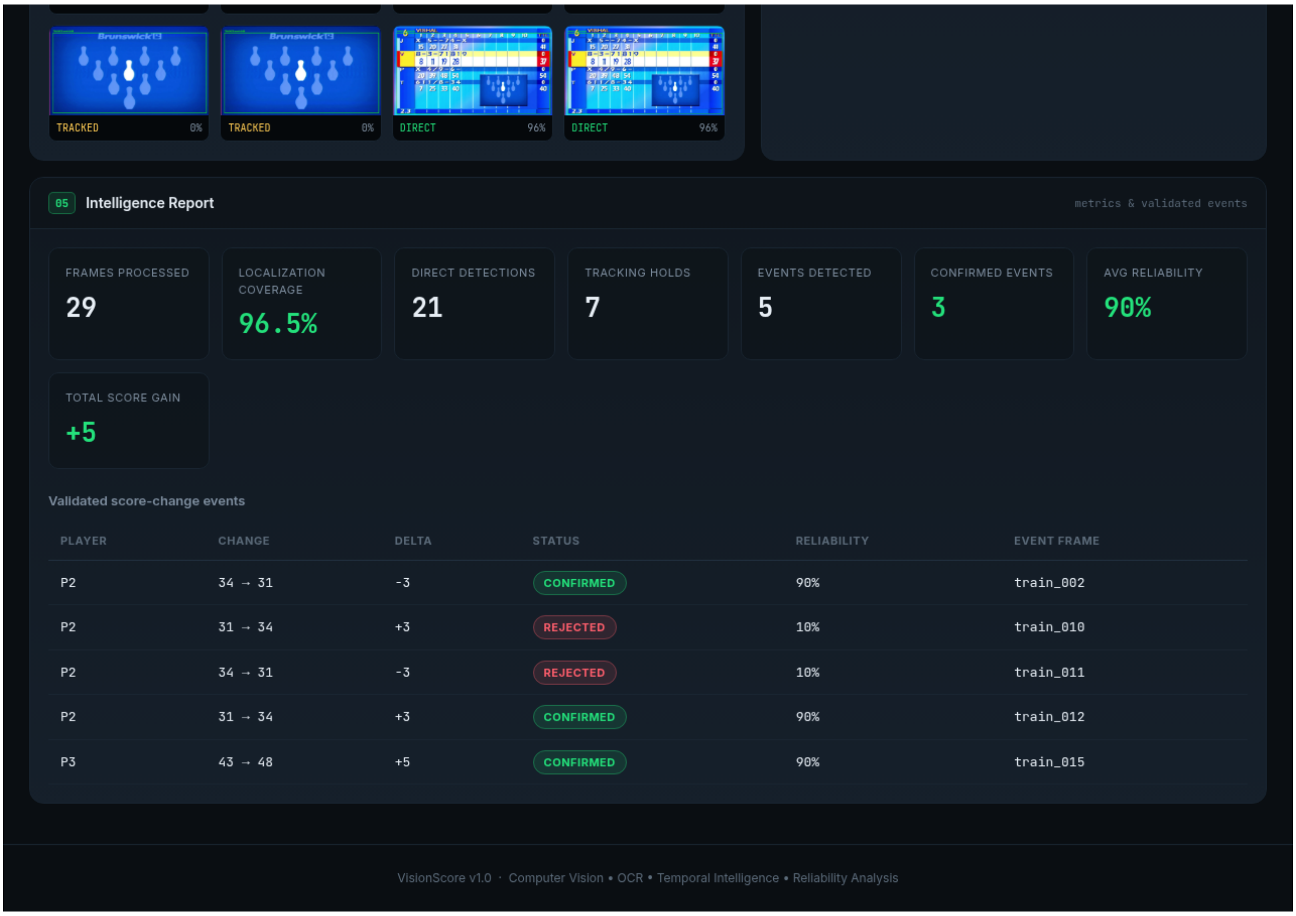
The full detection grid shows every frame's result side by side, giving a clear view of localization coverage across the whole clip.



Per-frame detection grid alongside the extracted data panel.

4. Extracted Scoreboard Data (Final Output)

The final output is the reliable per-player score, produced after temporal reasoning across all frames: **P1 = 20, P2 = 34, P3 = 48, P4 = 54**. The Intelligence Report summarizes the run - 29 frames processed, 96.5% localization coverage, and the validated score-change events.



Intelligence Report: metrics plus the validated events table. Note how the reliability layer marks stable changes CONFIRMED (90%) and OCR flickers REJECTED (10%).

Why this matters: a naive per-frame OCR reader would flip P2 to 31 on a misread frame. VisionScore rejects that flicker because the value reverts, and keeps the stable, correct result - turning noisy OCR into trustworthy data.